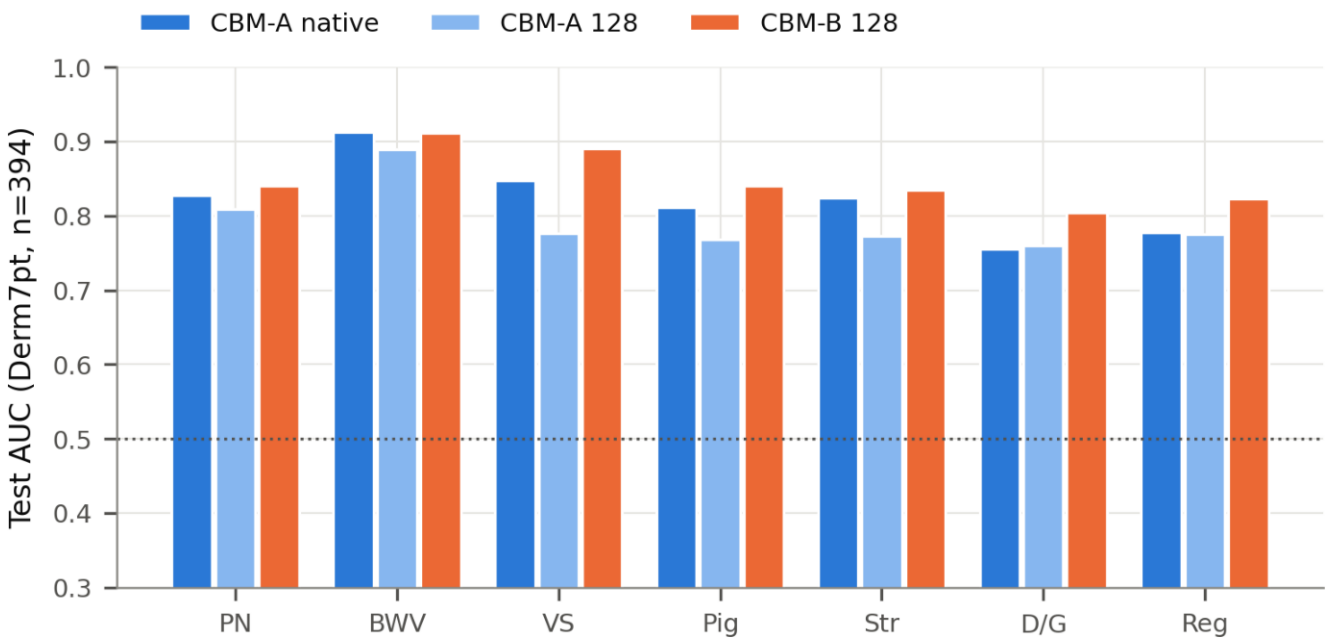
Supplementary Information

Testing concept-level audits of melanoma classifiers with diffusion counterfactuals and matched positive controls

Supplementary Figures S1 to S18. The Supplementary Tables are in Supplementary Data 1.

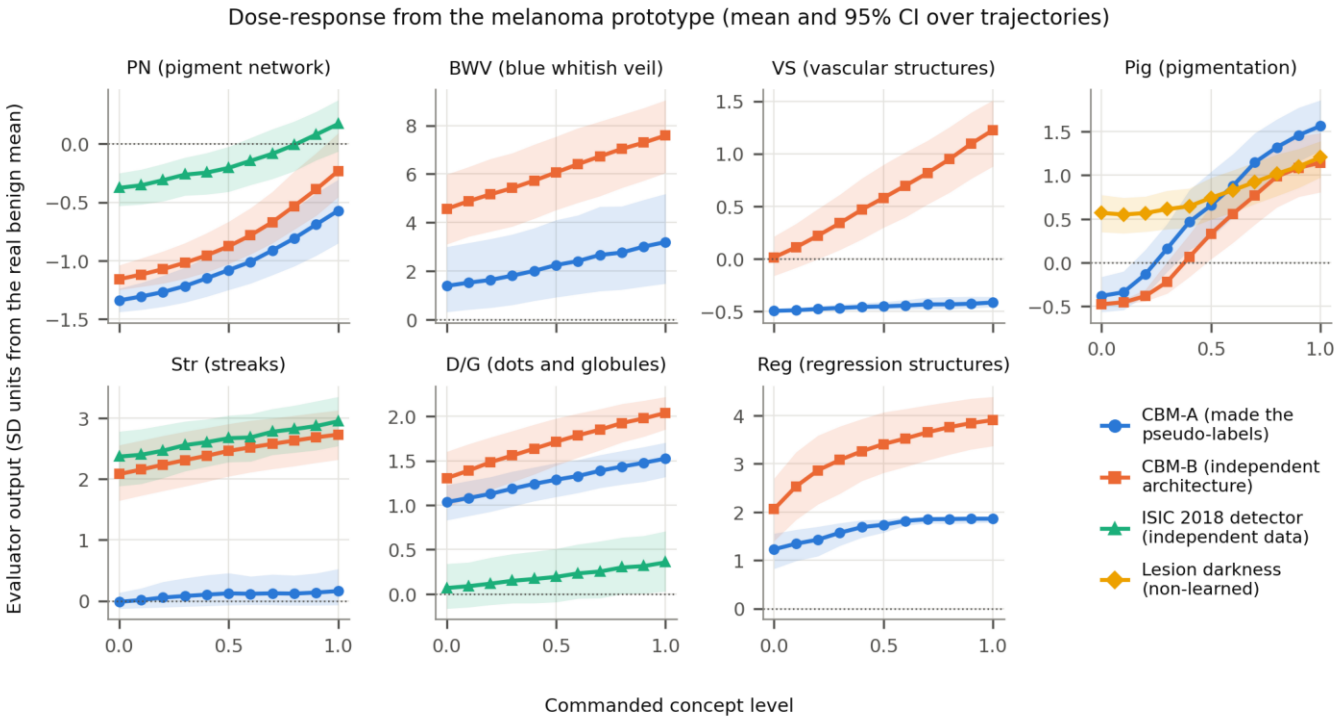
Supplementary Figure S1 | Evaluator AUC at native and 128-pixel input.

Per-concept test AUC on the official Derm7pt test split (n = 394) for CBM-A at its native resolution, CBM-A with the 128-pixel pipeline and CBM-B with the 128-pixel pipeline. Dotted line, chance.



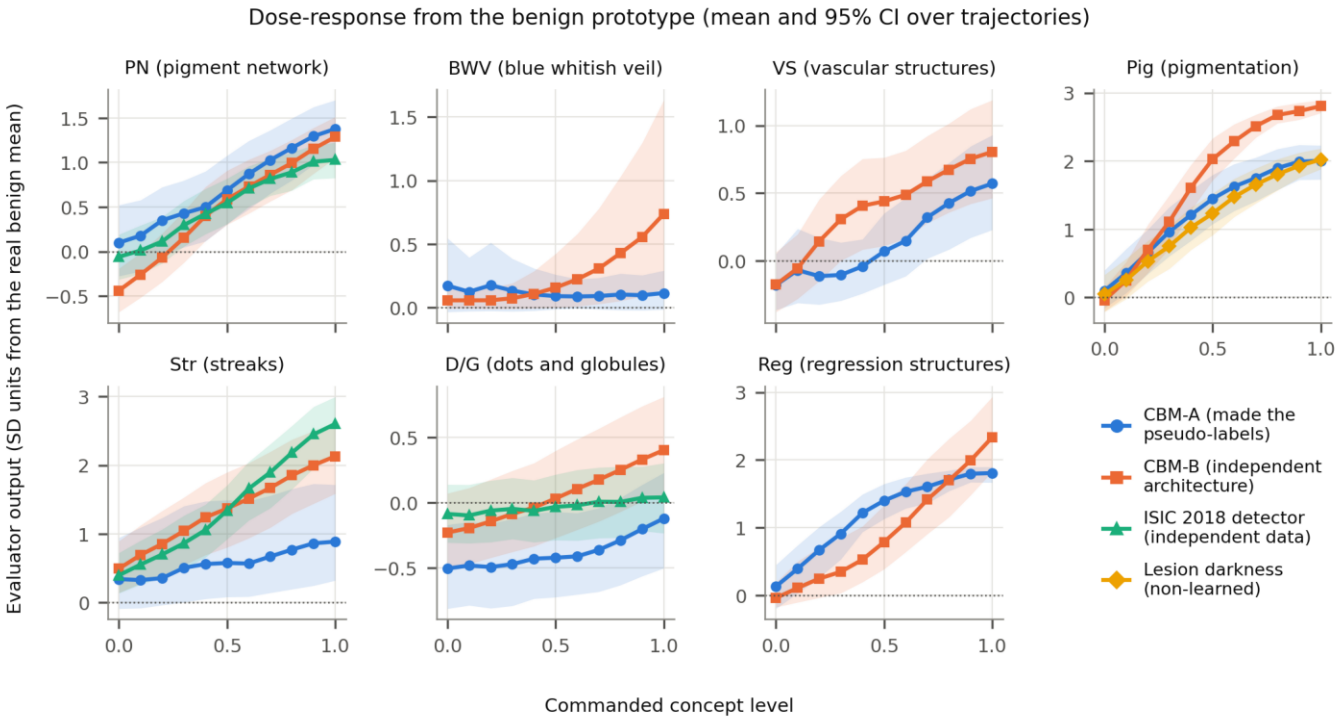
Supplementary Figure S2 | Dose responses on the melanoma prototype.

Evaluator outputs (SD units from the mean of real benign images) as each knob moves from 0 to 1 in steps of 0.1 on the melanoma prototype; mean and 95% interval over 32 trajectories. CBM-A made the pseudo-labels; CBM-B, the ISIC 2018 detector and lesion darkness are the other readings.



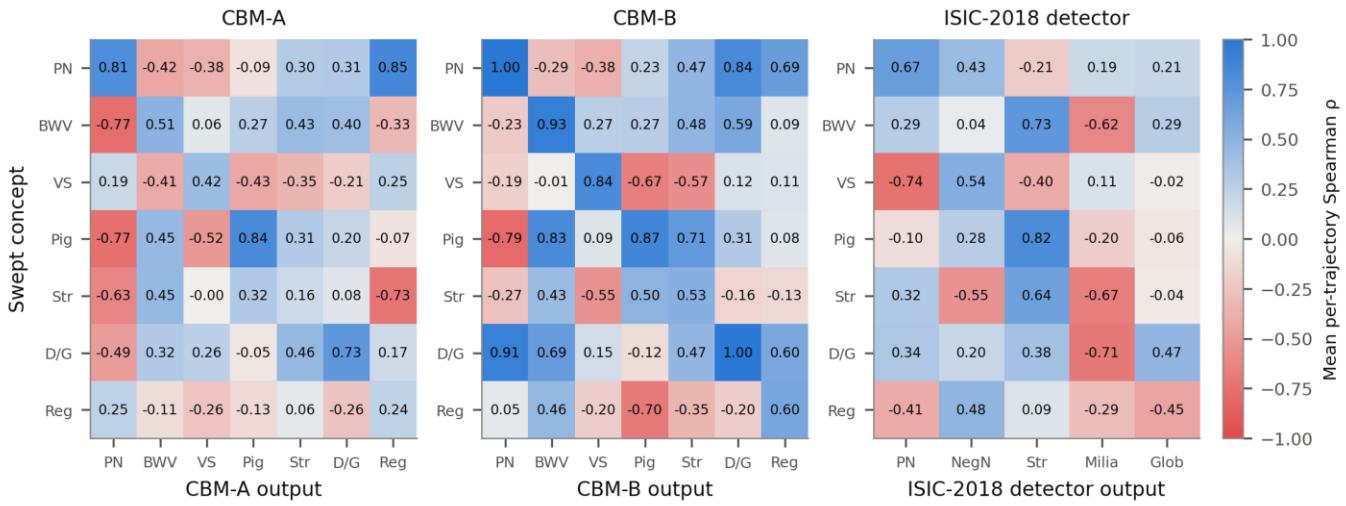
Supplementary Figure S3 | Dose responses on the benign prototype.

As Supplementary Figure S2, for the benign prototype.



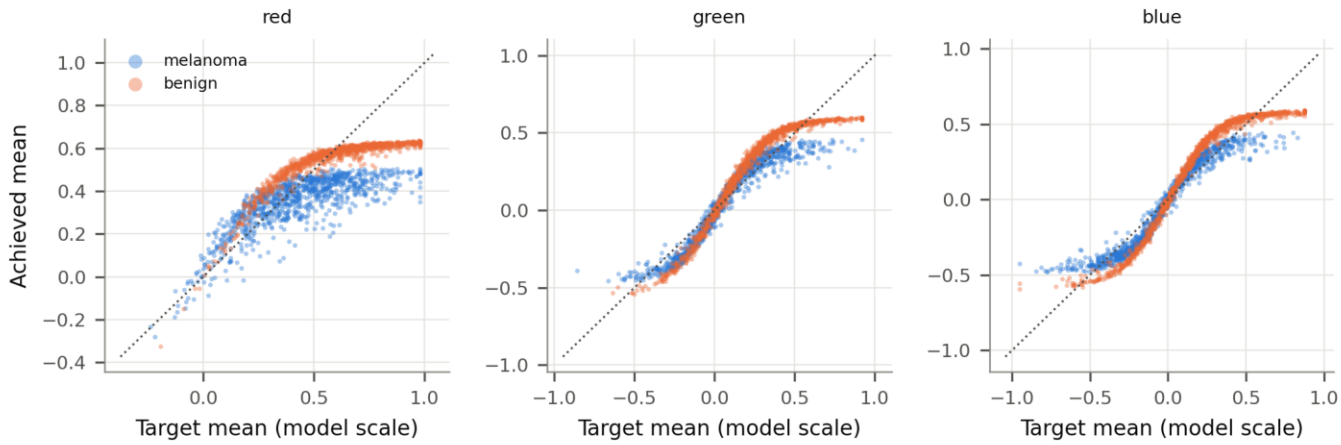
Supplementary Figure S4 | Knob cross-talk on the melanoma prototype.

Mean per-trajectory Spearman correlation between the level of the swept knob (rows) and each evaluator output (columns) for CBM-A, CBM-B and the ISIC 2018 detector. Off-diagonal values show changes in concepts other than the swept one.



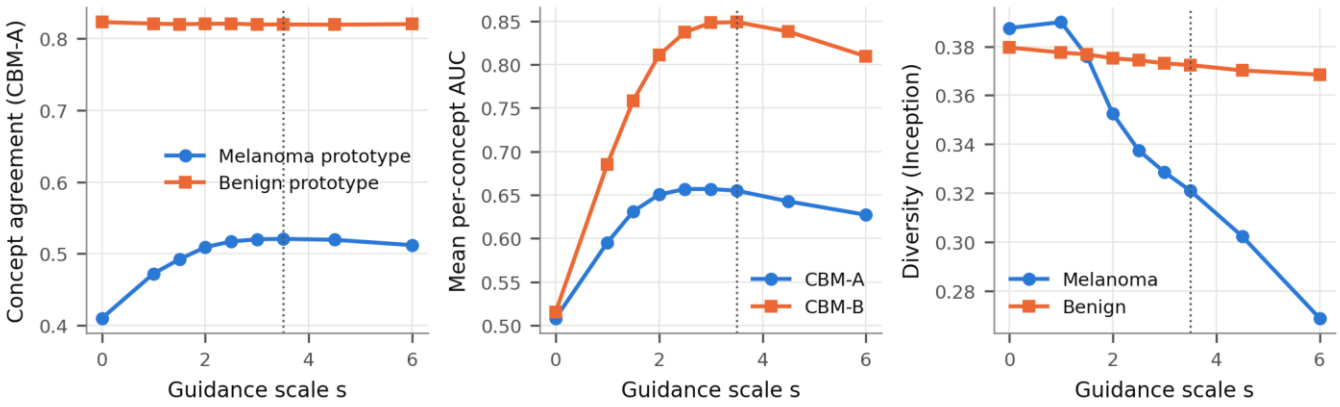
Supplementary Figure S5 | Colour targeting with the colour-calibrated start noise.

Achieved against targeted mean value per RGB channel (model scale) for generated melanoma and benign images. Dotted line, achieved equals target. The fit is weakest for the red channel.



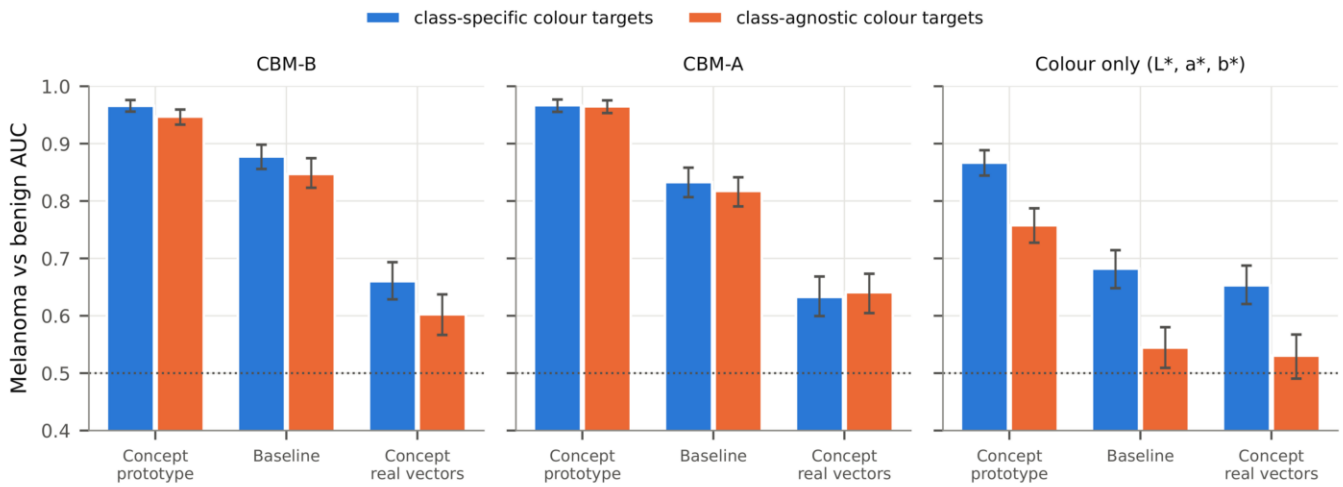
Supplementary Figure S6 | Guidance-scale sweep.

Concept agreement with CBM-A for the melanoma and benign prototypes (left), mean per-concept AUC of CBM-A and CBM-B on the generated images (middle) and Inception-feature diversity of generated melanoma and benign images (right), against the classifier-free guidance scale s. Dotted line, s = 3.5, the value used in all audits.



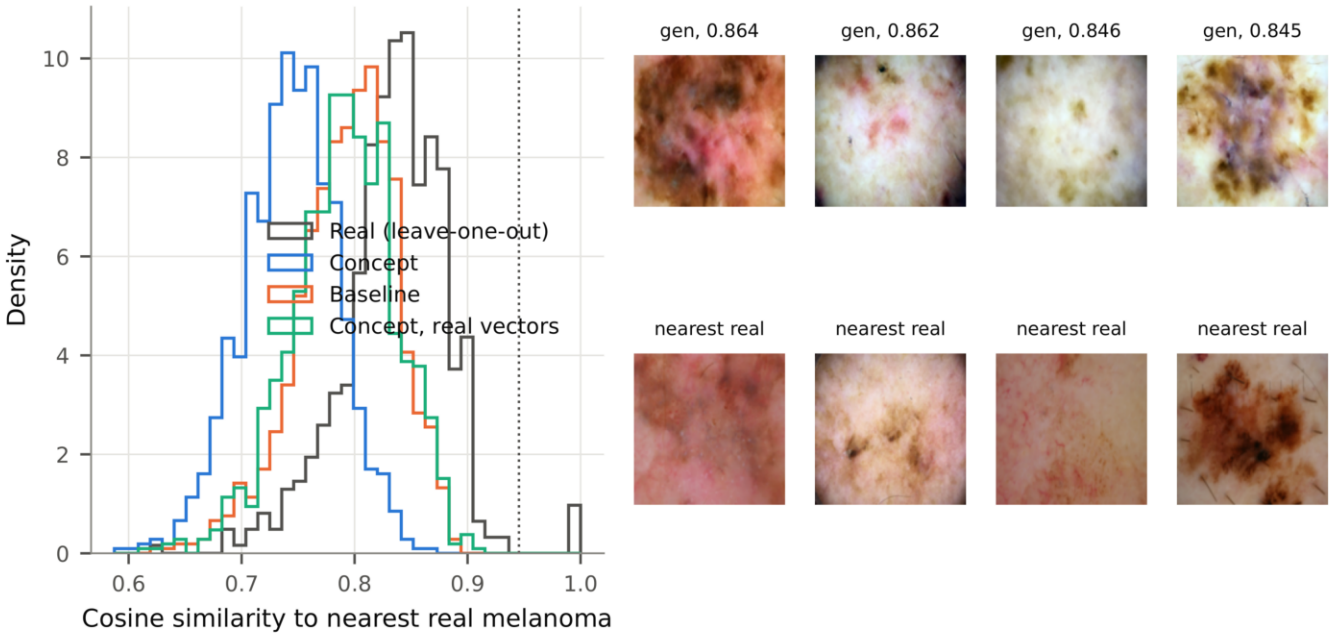
Supplementary Figure S7 | Class signal with class-specific and class-agnostic colour targets.

Melanoma-against-benign AUC of CBM-B, CBM-A and a colour-only classifier (mean L*, a*, b*) on generated images; error bars, 95% bootstrap intervals. Concept, concept-conditioned model; Baseline, class-conditioned model; prototype, class prototype vector; real vectors, pseudo-label vectors of real images.



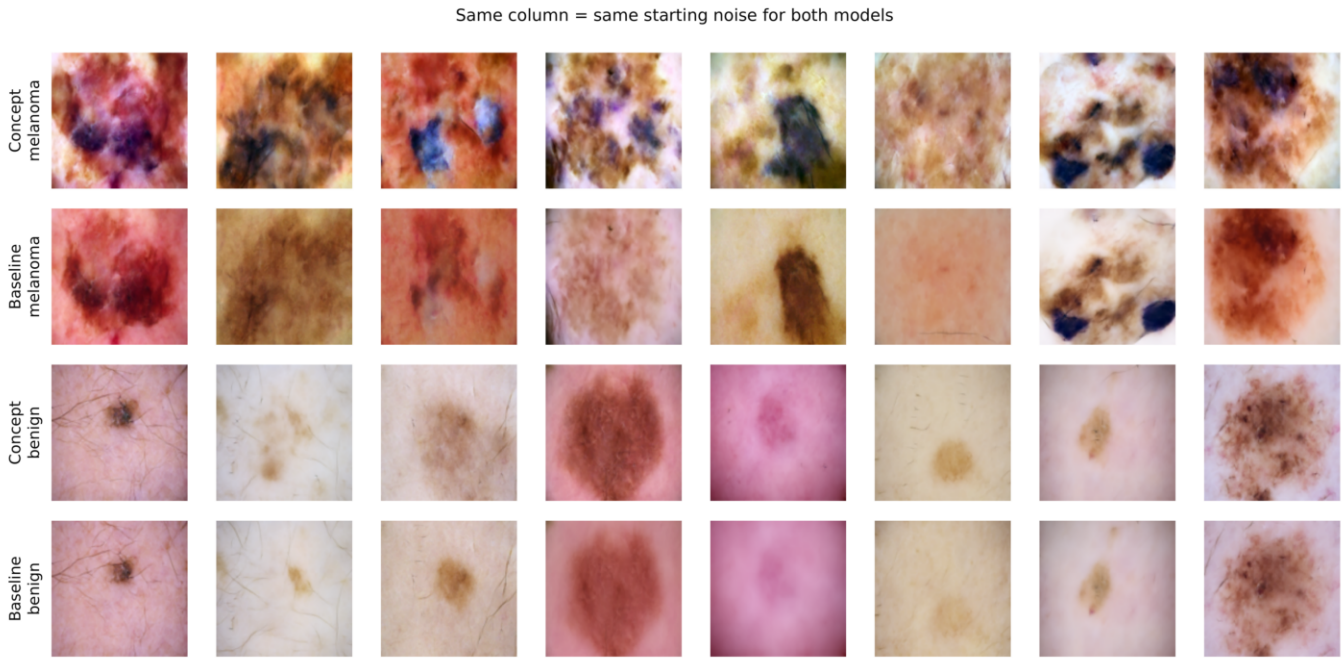
Supplementary Figure S8 | Memorisation screen.

Left, cosine similarity (Inception features) of each generated melanoma image to its nearest real training melanoma, with the real leave-one-out distribution as reference; dotted line, 99th percentile of the real distribution. Right, the four most similar generated images and their nearest real images. Concept, concept-conditioned model; Baseline, class-conditioned model; real vectors, pseudo-label vectors of real images.



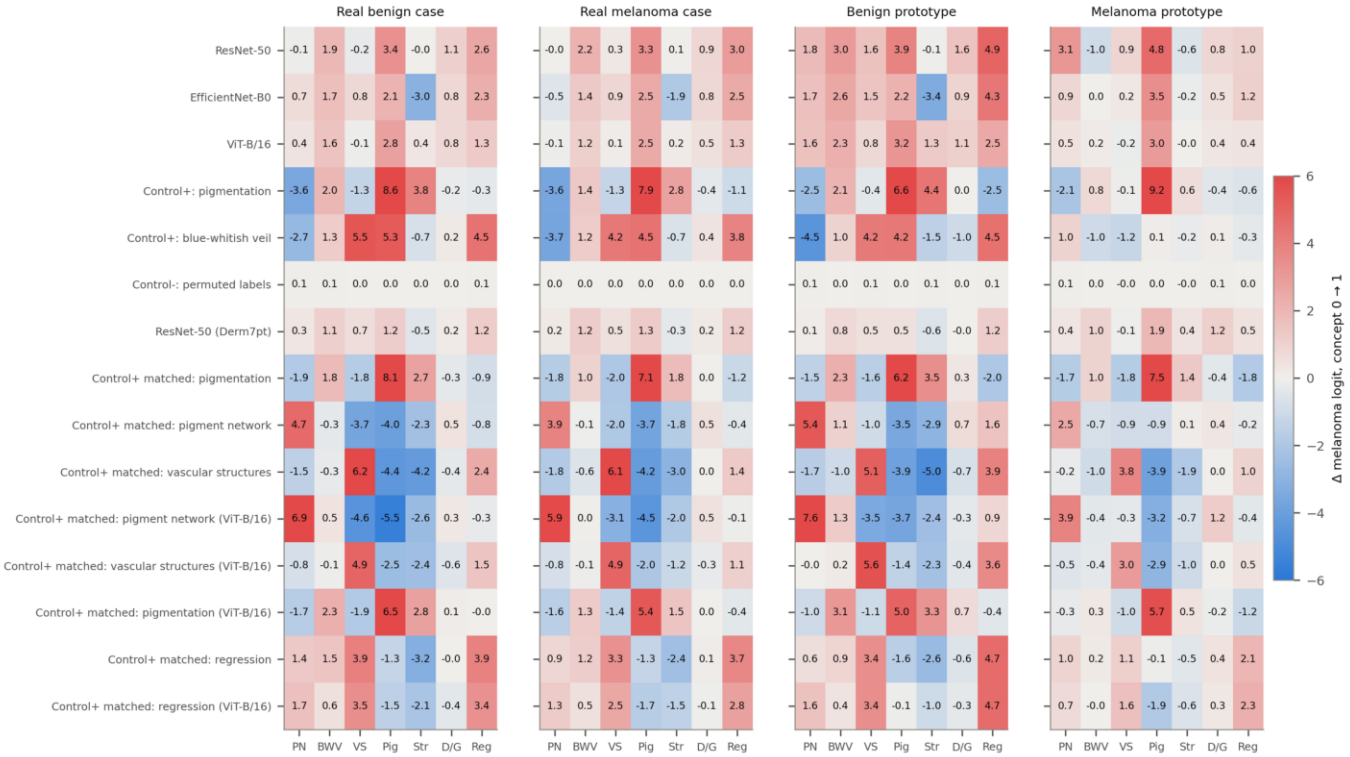
Supplementary Figure S9 | Paired qualitative examples.

Generated melanoma and benign images from the concept-conditioned (Concept) and class-conditioned (Baseline) models; images in one column share the start noise.



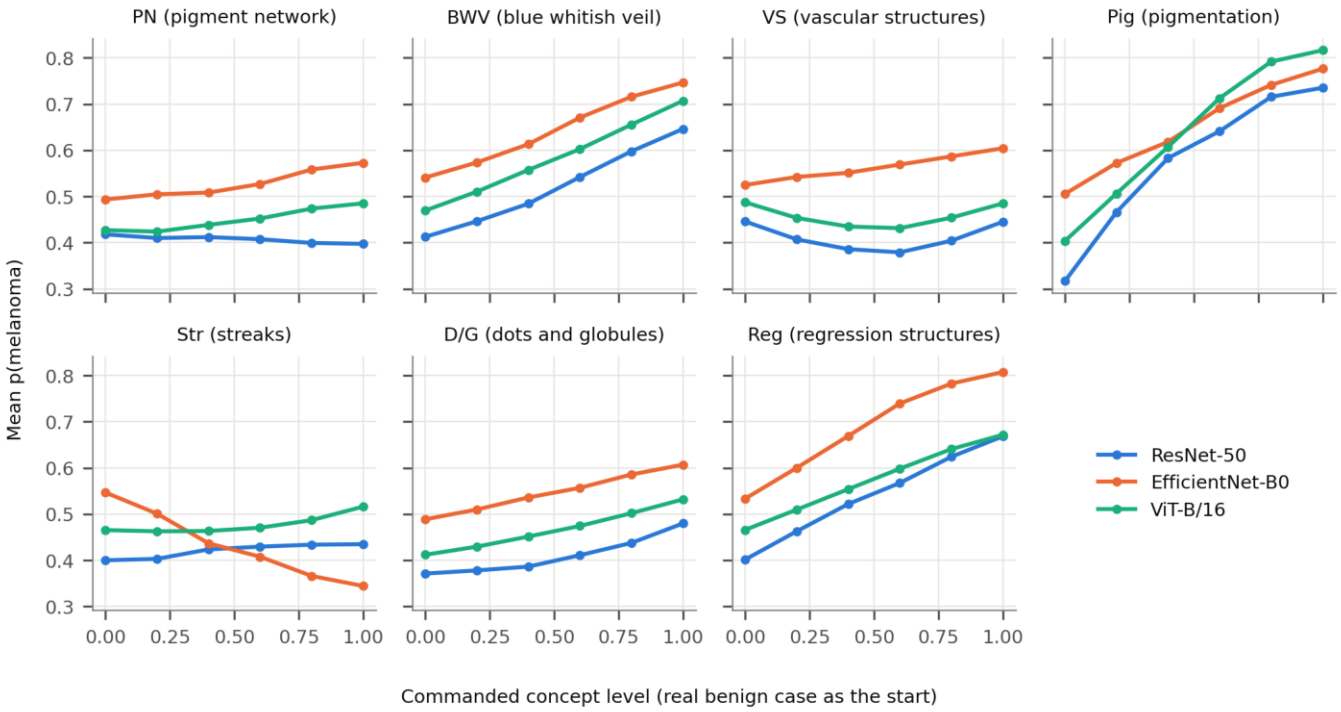
Supplementary Figure S10 | Audit effects on all four bases.

Change in the melanoma logit from knob level 0 to 1 for the classifiers named on the left, on the two realistic bases and the two prototypes (mean over trajectories).



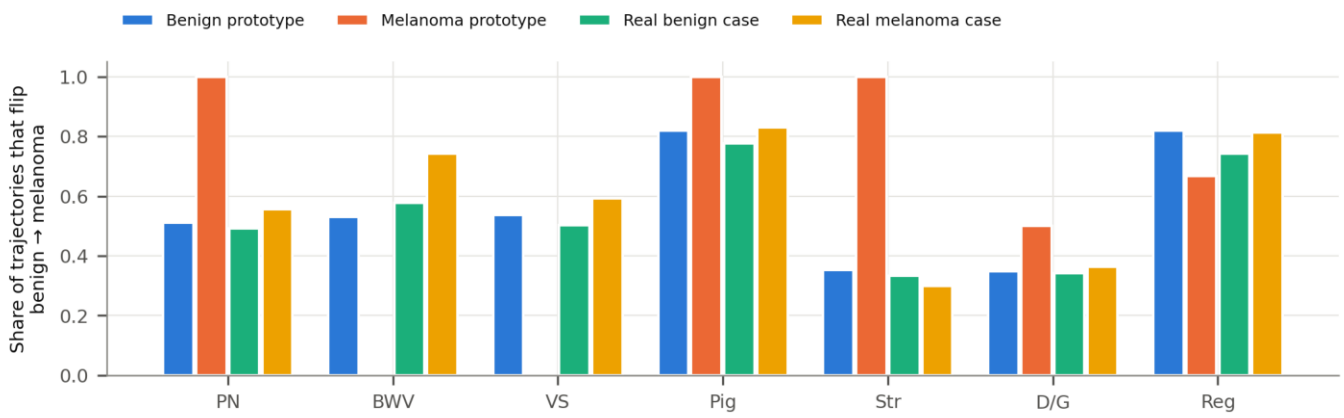
Supplementary Figure S11 | Dose responses on the real benign base.

Mean p(melanoma) over trajectories against the knob level for each architecture (realistic benign base, 6 levels). The knobs change other concepts as well (Fig. 2c).



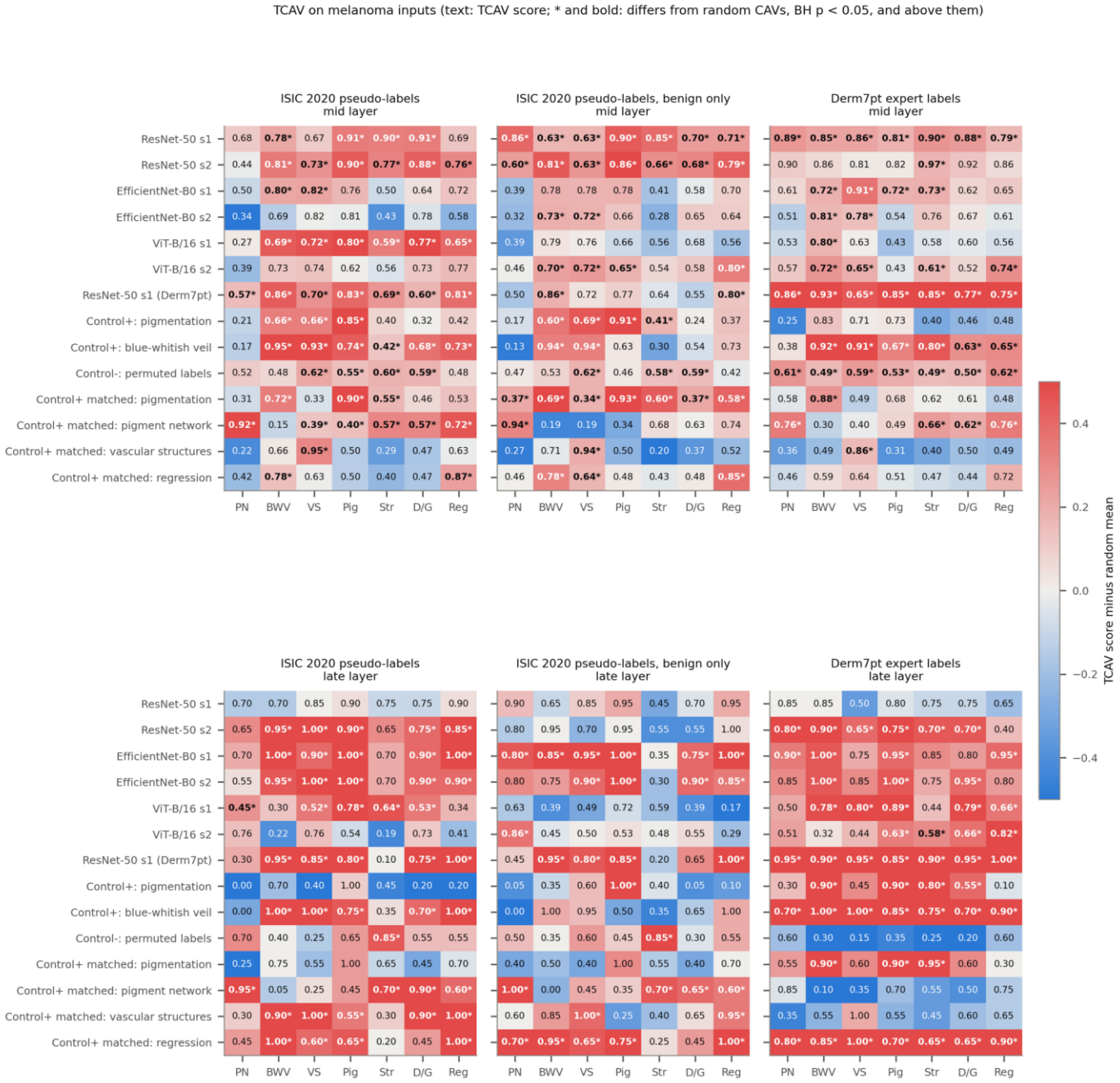
Supplementary Figure S12 | Decision flips.

Share of trajectories whose decision changes from benign to melanoma between knob level 0 and 1, at the threshold that gives 90% sensitivity on the validation set, for each knob and base.



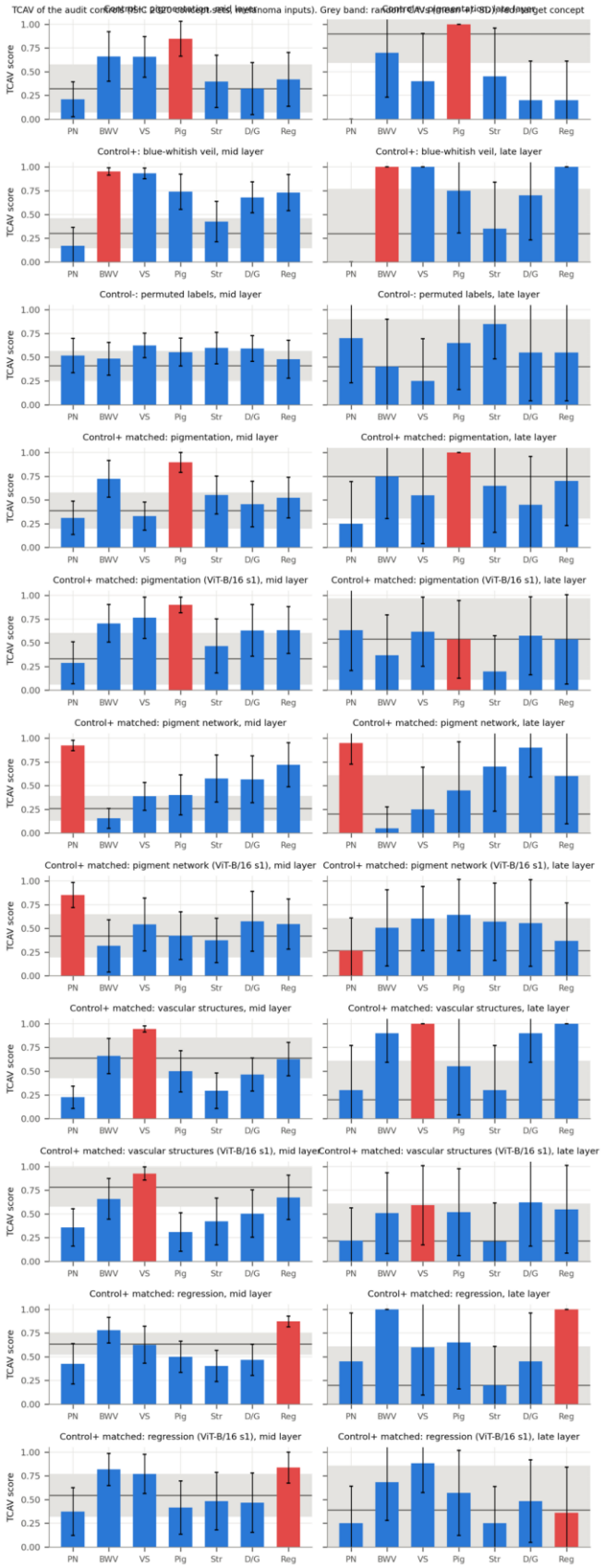
Supplementary Figure S13 | TCAV scores of all classifiers (our implementation).

Mean TCAV score on melanoma target images for the middle and last layers and the three concept-set types; colour, score minus the random-CAV mean; asterisk and bold, different from random CAVs (Benjamini-Hochberg p < 0.05) and above them.



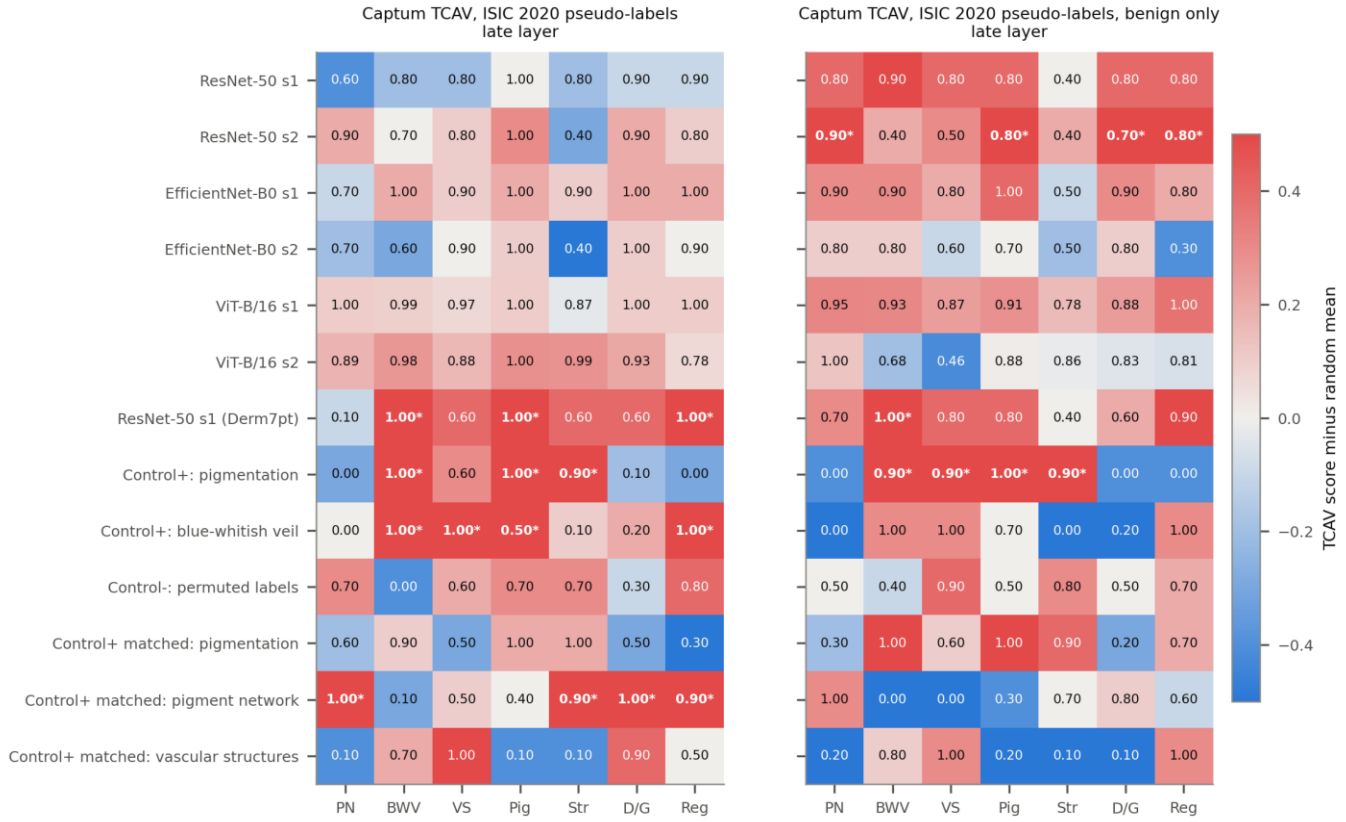
Supplementary Figure S14 | TCAV results for the controls.

TCAV score per concept (mean and SD over runs) for each control at the middle and last layers, with ISIC 2020 concept sets from all training images and melanoma inputs; red, the target concept; grey band and line, random CAVs (mean and SD).



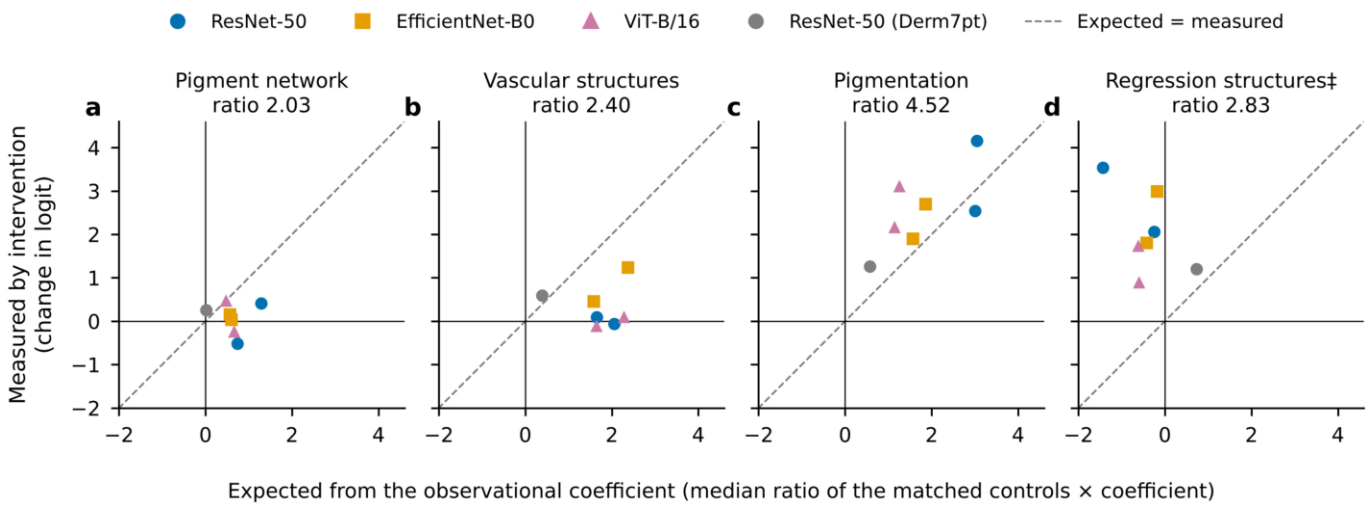
Supplementary Figure S15 | Captum TCAV scores.

Captum 0.9.0 TCAV scores at the last layer for the two ISIC concept-set types; colour, score minus the random mean; asterisk and bold, different from random sets.



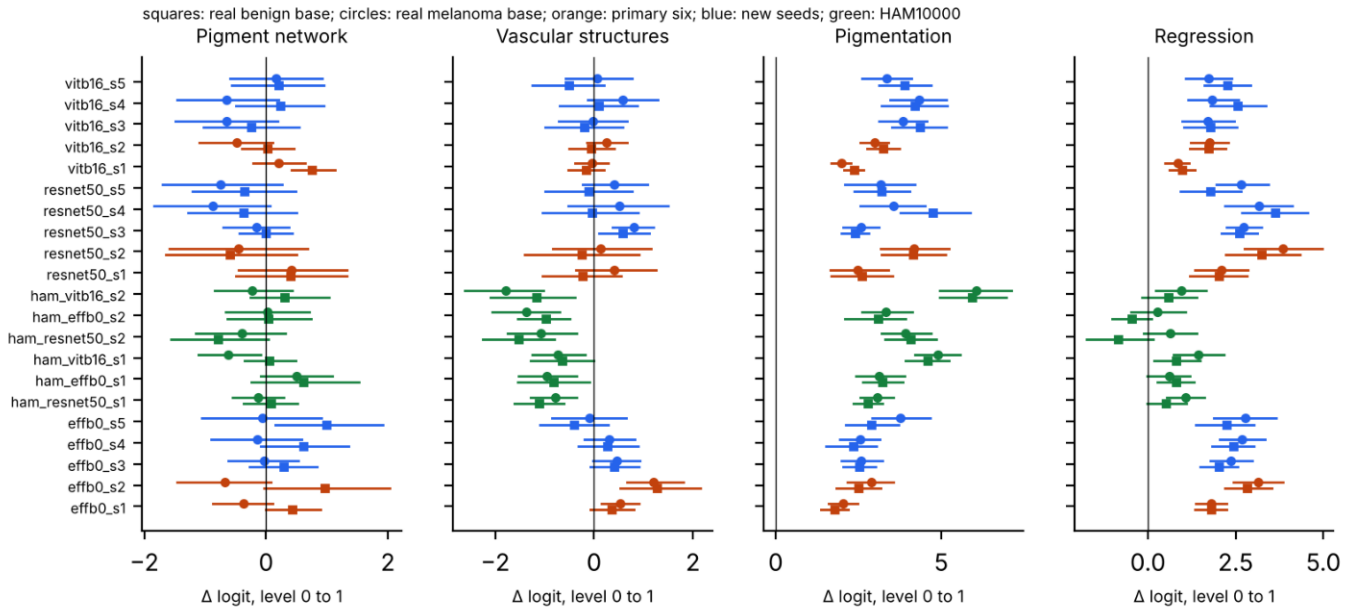
Supplementary Figure S16 | Observational prediction against the measured intervention (exploratory).

For each target-dominant knob, the x axis is the audit effect that the observational coefficient predicts (coefficient x median ratio of audit effect to coefficient in the four CBM-A matched controls; ratio in the panel title). The y axis is the measured audit effect (mean of the two realistic bases). Dashed line, prediction equals measurement. Defined after the results; it does not remove knob cross-talk.



Supplementary Figure S17 | Sweep effects of every melanoma classifier on each realistic base.

Change in the melanoma logit from knob level 0 to 1 with 95% bootstrap intervals over trajectories, for the six primary ISIC 2020 classifiers (orange), the nine retrained ISIC 2020 classifiers (blue, seeds 3 to 5) and the six HAM10000 classifiers (green). Squares, real benign base; circles, real melanoma base.



Supplementary Figure S18 | HAM10000 near-duplicate screen.

The 12 HAM10000 images (top) with the highest cosine similarity of ImageNet ResNet-50 features to an ISIC 2020 classifier image (bottom); similarity above each pair. No pair reached 0.98. On visual review, each pair shows different lesions; much of the similarity comes from hair and skin texture. Report only; no image was removed.

